



SEGURANÇA

Project 02	Deadline: 2026/03/05 (12:59)	2026/02/26
Expected time: 90 minutes	Non-contact hours	
Name: _____	N.: _____	Total.: _____

Este projeto deve ser realizado até à data acima mencionada (deadline). Deve fazer um relatório resumido com screenshots e texto que os enquadre, que evidencie e contextualize a sua realização. Deve usar a mesma numeração de perguntas e identificação das alíneas das perguntas para as respostas. Anote na capa do relatório o nome da unidade curricular (Segurança), o ano letivo (2025/2026) o seu nome completo, email, número de aluno, data de entrega e o tempo que demorou a realizar o projeto. Dentro de uma pasta "Project02" coloque o relatório em PDF assim como os ficheiros das capturas de tráfego relevantes. O nome de cada ficheiro de captura de tráfego deve começar pelo número e alínea da pergunta em causa (ex. 4p-Telnet.pcap). Comprima em formato zip a pasta com todos os ficheiros (Project02.zip) e entregue o ficheiro na respetiva atividade do Nónio. O projeto poderá ainda ser avaliado em aula por teste e/ou inspeção (topologia funcional e/ou capturas de tráfego). Expressões neste enunciado a azul e sublinhadas representam ligações a recursos externos. Leia o enunciado até ao final antes de iniciar a realização do projeto. Reserve um período um pouco mais extenso que o tempo previsto para realização do projeto em que seja garantido não ser interrompido. Faça pausas a cada 40 minutos.

1. Clone the topology you developed during Project 01. Ensure that GNS3 does not reassign different mac addresses. Ensure you strictly respect all the instructions provided on point 1 and 2 of the previous Project. Rename R1 (using the `hostname` command) as R1-NNNN, where NNNN is the last 4 digits of your student number. (Conserve without mutations all the projects you develop as well as the related resources until you complete the Security course. Those materials can be requested during any lesson and can also be used by you during project assessments.)
2. This project assumes that you have already studied the first four modules of the [CCNA Security](#) course. You must take the preparatory exam (*Modules 1 - 4: Securing Networks Group Exam*) covering such content and review the subjects needed. You should also read this [Cisco Guide to Harden Cisco IOS Devices](#) on your preferred format (also available on Nónio).
3. In this new project you will improve the base configuration previously performed and observe several relevant facts regarding the insecure nature of the TCP/IP protocol stack.
4. Regarding module 4 (*Secure Device Access*), perform the following tasks, strictly following the instructions provided. You should take notes on a specific document regarding your observations. This will help you to speed up the in-person assessment of this project.
 - a. From time to time save the current router configuration to its flash using the command `copy running-configuration startup-configuration`. This way the configuration becomes non-volatile and becomes available on the next router's boot.
 - b. At R1 run the following command `show ip interface brief` and screenshot its output. Check if all the interfaces are up and well configured. Use the `ping` command to check IP reachability between A1 and R1.
 - c. Define a minimum password length of 10 characters on R1.
 - d. Protect the "user EXEC mode" of R1 with a password for out-of-band management purposes.
 - e. Run the `ifconfig eth0` command at the A1 terminal. Try to access the R1 router from it using the `telnet IP(R1.e0/0)` command. Screenshot the terminal output so that you capture both outputs. Do you already have remote access to the router? Why?
 - f. Protect the in-band management access (covering all the VTY lines available) with a password. Enable `telnet` access to it (just `telnet`).
 - g. Attempt to remotely access the router from A1 again. Do you have access to the router?
 - h. Protect the "privileged EXEC mode" with a password. Do you have access to the router?
 - i. Inspect the current configuration (`show running-configuration`) and check how the passwords are stored. Are they stored in cleartext? Take a screenshot of the current relevant configuration and comment on.
 - j. Activate the password encryption service of IOS. Inspect again the running configuration and screenshot the current relevant configuration related with the protection you are working on. Discuss the output.

- k. Search for sites that allow to decrypt such weak Cisco encryption. Try one of the sites with the encrypted password of the router's console line. Screenshot the browser window so that the URL of the website becomes visible, as well as the decrypted password. Read [this document](#) to learn about the Cisco position regarding this sensible subject.
 - l. Repeat the procedure with the encrypted password used to protect the "privileged EXEC mode". Do you succeed reverting the encrypted password to cleartext this time? Why? Do you know that even the way the "enable password" is stored (actually, the password is not stored on IOS, but instead a hash of it) is nowadays considered weak since the hash is built with the MD5 algorithm? Learn more about this on [RFC 6331](#).
 - m. The MD5 hash used by Cisco includes some "salt" to be stronger. If you delete the command and define the same `enable secret` you will get a different hash value. This is why you cannot just copy and paste the value you get from the `show running-config` command. Even so the overall process is weak since MD5 is currently considered weak. Check it yourself: imagine a simple weak password, get its MD5 hash using this [site](#). Then try to revert the hash obtained using this [site](#). If it does not work repeat the process with another weak password. Report the experiment with screenshots.
 - n. Read a more detailed perspective [here](#). Find out the current best option [here](#). Correct your configuration so it becomes more secure. Use the more complete command `enable algorithm-type ...` this time. Screenshot the command line you provide.
 - o. Read about the `/etc/shadow` format ([manpage](#) and [summary](#)). Edit this file on one of your own Linux VMs and check the contents.
 - p. Quit the `telnet` session from A1 to R1. Initiate a capture session with Wireshark between A1 and R1. Setup the visualization filter so that you see just the traffic related to the `telnet` session. You can check the [IANA register](#) to discover the transport protocol and the port `telnet` uses. After the Wireshark sniffer is running with the filter establish a `telnet` session from A1 to R1. After you login correctly enable the privilege EXEC mode. Stop the capture and save it to your project resources. Screenshot the summary of the first 10 PDUs captured during the telnet session. Analyze and comment on. On Wireshark select the first PDU from the telnet session, open the Analyze menu and select the Follow option. Screenshot the window. Analyze and comment on. Is the telnet protocol secure to (remotely) manage systems?
 - q. Improve the configuration so that more than three failed telnet login attempts within 30 seconds trigger the lockout of the VTY lines for 90 seconds. Run the command `date` on A1, then perform a brute force attack to R1 from A1 with telnet (i.e., using a wrong password three times within a 30 second period). Run `date` at end. Telnet R1 again. Wait two minutes. Run `date` again. Telnet R1 with the correct password. Screenshot the terminal output of A1 so that all the experiment becomes visible.
 - r. Make the R1 logs every authentication attempt. Remote access R1 using telnet and screenshot the log message it triggers as well as the output of the command `show login`.
 - s. Define banners to all possible situations. Some text must alert users about related legal issues. Check sites like [this one](#) if you like a bit of art-work. Screenshot the banner you define to be visible at the beginning of a remote telnet access.
 - t. We will focus SSH on the next practical lesson as well as Access Control Lists (ACLs), so you can avoid the Denial of Service (DoS) condition that a brute force attack against the telnet password can trigger.
5. Turn your running configuration into saved configuration.